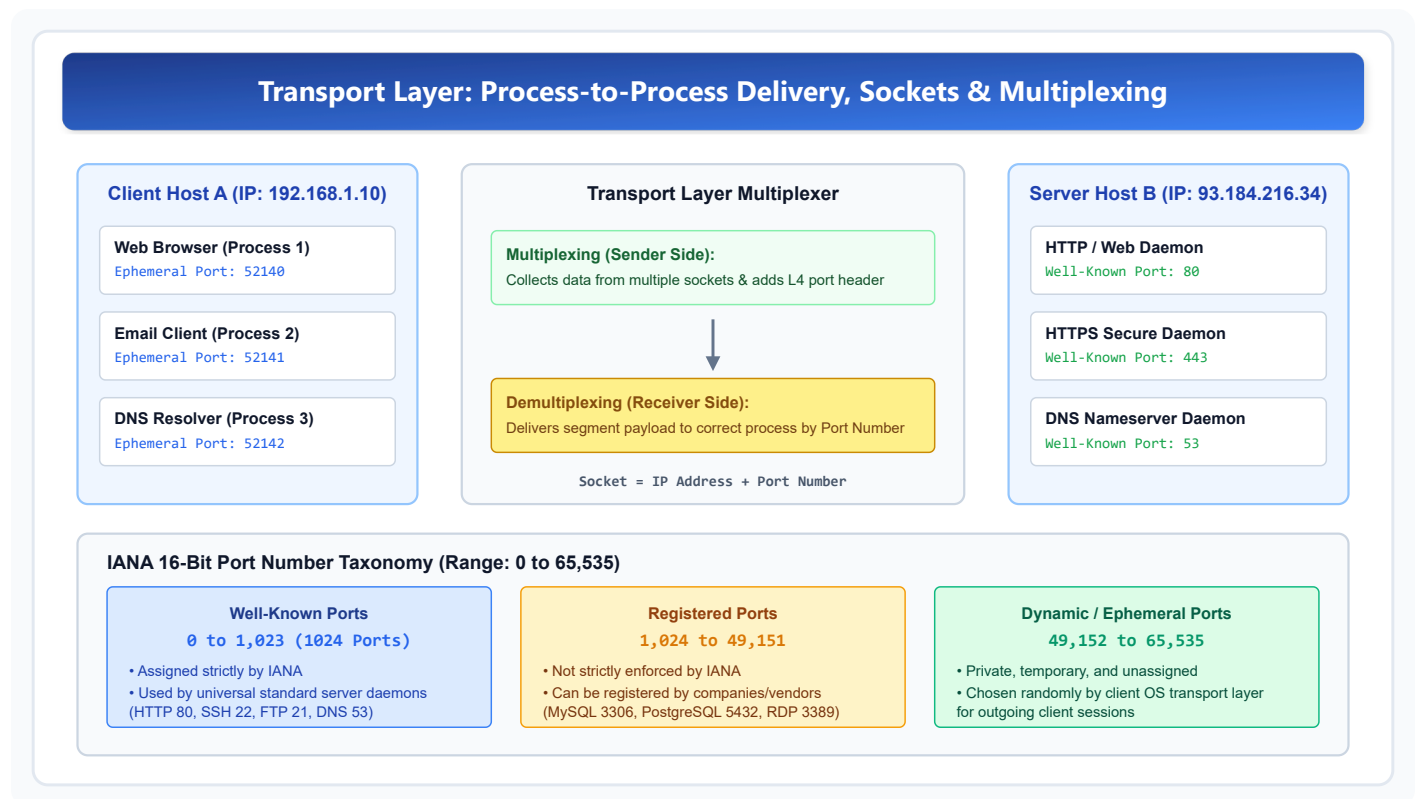


Module 01: Transport Layer Services and Port Addressing

Unit 4: Transport Layer

Process-to-Process Delivery

End-to-End Communication Mechanics, 16-Bit Port Hierarchy, Socket Addresses, Multiplexing vs Demultiplexing, and Connection Paradigms.



1. Port Number Taxonomy (IANA Standard)

Well-Known Ports: 0 to 1,023 (Universal servers: HTTP 80, DNS 53, SSH 22)

Registered Ports: 1,024 to 49,151 (Vendor daemons: MySQL 3306, RDP 3389)

Dynamic / Ephemeral Ports: 49,152 to 65,535 (Temporary client ports chosen by OS)

Socket Address = IP Address : Port Number (e.g., 192.168.1.10:52140)

2. 5-Tuple Connection Uniqueness

Every active connection across the Internet is uniquely differentiated by the 5-Tuple: **{Source IP, Source Port, Dest IP, Dest Port, Protocol}**.